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COP5615 — DISTRIBUTED OPERATING SYSTEMS PRINCIPLES

Project 2

Gossip Simulator

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Abstract

This project implements asynchronous Gossip and Push-Sum simulations using Erlang actors. Four network topologies are supported: full, line, two-dimensional grid, and imperfect two-dimensional grid. A centralized statistics actor collects convergence information and records elapsed time. Baseline experiments evaluate both algorithms across network sizes from 4 to 49 nodes. An additional 360-trial experiment investigates how changing the local Gossip reception threshold affects convergence. In the baseline experiments, all 72 Push-Sum trials succeeded, whereas 39 of 72 Gossip trials succeeded. The threshold experiment demonstrates that increasing the local reception threshold substantially improves observed Gossip success, particularly for larger and less-connected networks.

1 Introduction

The objective is to simulate distributed information propagation using independent Erlang actors communicating through asynchronous messages. Each actor represents a node in a network and interacts with its neighbors according to the selected algorithm. The simulator accepts a node count, topology, and algorithm through the command-line interface:

```
project2 numNodes topology algorithm
```

Supported topologies are `full`, `line`, `2D`, and `imp2D`. Supported algorithms are `gossip` and `push-sum`. The experiments use network sizes $N \in \{4, 9, 16, 25, 36, 49\}$. The grid-based topologies use perfect-square node counts.

2 Implementation

2.1 Actor architecture

Each simulated node runs as an Erlang process. Actors communicate using messages rather than shared mutable state. The implementation separates actor behavior, topology construction, coordination, and statistics collection. The coordinator builds the topology, starts the actors, waits for their registration, configures their neighbors, initiates the selected algorithm, and waits for the convergence result. A centralized statistics actor collects actor registrations, algorithm updates, completion notifications, and convergence information. Elapsed time is measured with Erlang's monotonic clock. The recorded duration is the elapsed time between algorithm initiation and the detected convergence event, expressed in microseconds.

2.2 Gossip

Gossip begins when the coordinator sends a rumor to one actor. An actor that receives the rumor increments its local reception count. While it remains active, it randomly selects a neighbor and forwards the rumor. Once its local reception count reaches the configured threshold, the actor reports completion and stops forwarding the rumor. The baseline experiments use the default threshold of 10 receptions. The additional threshold experiment evaluates thresholds of 10, 20, 50, 100, and 500. The threshold is a **local reception count per actor**; it is not a global number of communication rounds. Because actors operate asynchronously and select neighbors randomly, a larger threshold can allow information to continue propagating for longer before individual actors stop forwarding it.

2.3 Push-Sum

Each actor begins with its node identifier as its s value and $w = 1$. The actor's current estimate is the ratio s/w. During communication, an actor divides its current s and w values by two and sends half to a selected neighbor, retaining the other half. The implementation uses an epsilon of 10^{-10} and requires three consecutive stable rounds, as defined by the implementation, before an actor reports that its estimate has converged. The coordinator and statistics actor collect the resulting completion information.

2.4 Network topologies

- **Full:** each node is connected to all other nodes.
- **Line:** nodes are arranged in a linear chain.
- **2D grid:** nodes are arranged in a square grid, with neighbor relationships determined by grid adjacency.
- **Imperfect 2D grid:** a two-dimensional grid is augmented with an additional random neighbor connection.

3 Experimental Methodology

The baseline dataset contains 144 trials: two algorithms, four topologies, six network sizes, and three trials per configuration. The Gossip threshold dataset contains 360 trials: four topologies, six network sizes, five thresholds, and three trials per configuration. A trial is counted as successful when the experiment harness receives a successful convergence result. Trials without a successful result are counted as failures. The coordinator uses a 30-second convergence wait in the current implementation. Therefore, a failed trial means convergence was not detected successfully within the experiment's execution conditions; it does not establish that convergence is mathematically impossible. For successful trials, the analysis script computes the mean convergence time and, when at least two successful observations are available, the sample standard deviation. Failed trials are included in the success-rate denominator but excluded from convergence-time averages. The experiments use only three trials per configuration. The results should therefore be interpreted as observations of this simulator under the tested conditions, rather than precise estimates of long-run success probabilities.

4 Baseline Results

4.1 Overall outcomes

Table 1 summarizes the baseline success rates by algorithm and topology. Each entry represents 18 trials: six network sizes with three trials per size.

Table 1: Baseline convergence outcomes by algorithm and topology.

Algorithm	Topology	Successful	Failed	Success rate
Gossip	Full	16	2	88.9
Gossip	Line	7	11	38.9
Gossip	2D	7	11	38.9
Gossip	Imperfect 2D	9	9	50.0
Push-Sum	Full	18	0	100.0
Push-Sum	Line	18	0	100.0
Push-Sum	2D	18	0	100.0
Push-Sum	Imperfect 2D	18	0	100.0

Push-Sum succeeded in all 72 baseline trials, including every 49-node configuration. Gossip was more sensitive to topology and network size. The full topology had the highest Gossip success rate, while the line and 2D topologies had the lowest rates in this baseline dataset.

4.2 Convergence time

Figures 1 and 2 show the mean convergence times of successful baseline trials as network size increases. These means exclude failed trials, so a plotted point does not represent the expected time to completion across all attempts.

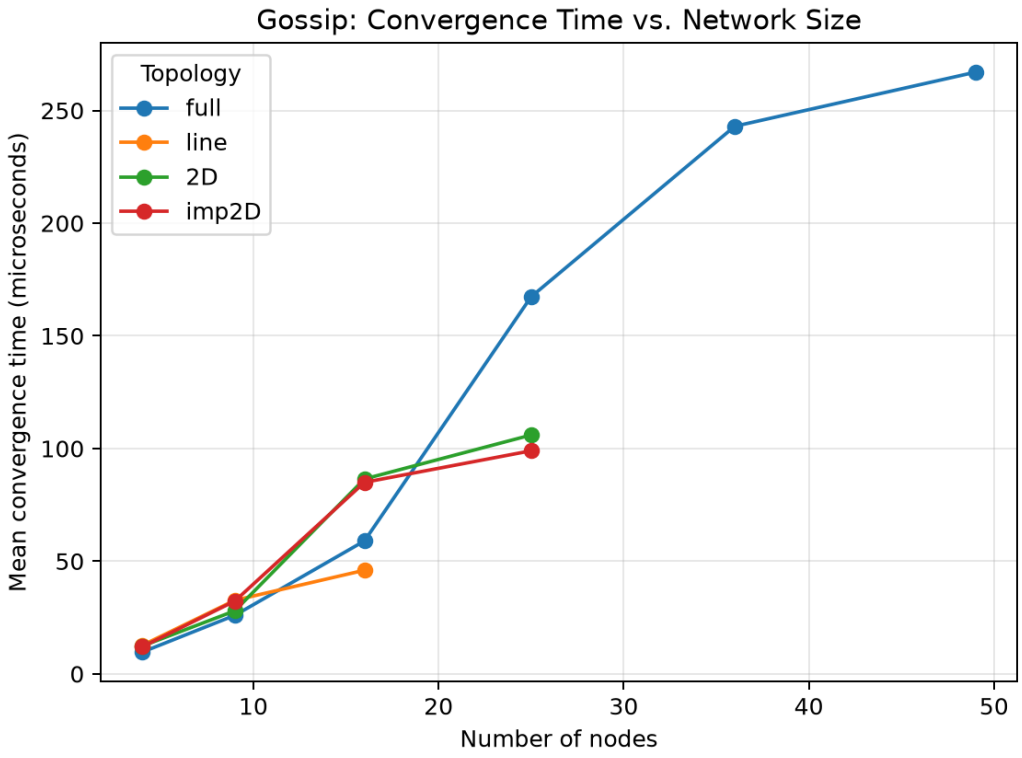


Figure 1: Mean Gossip convergence time for successful baseline trials.

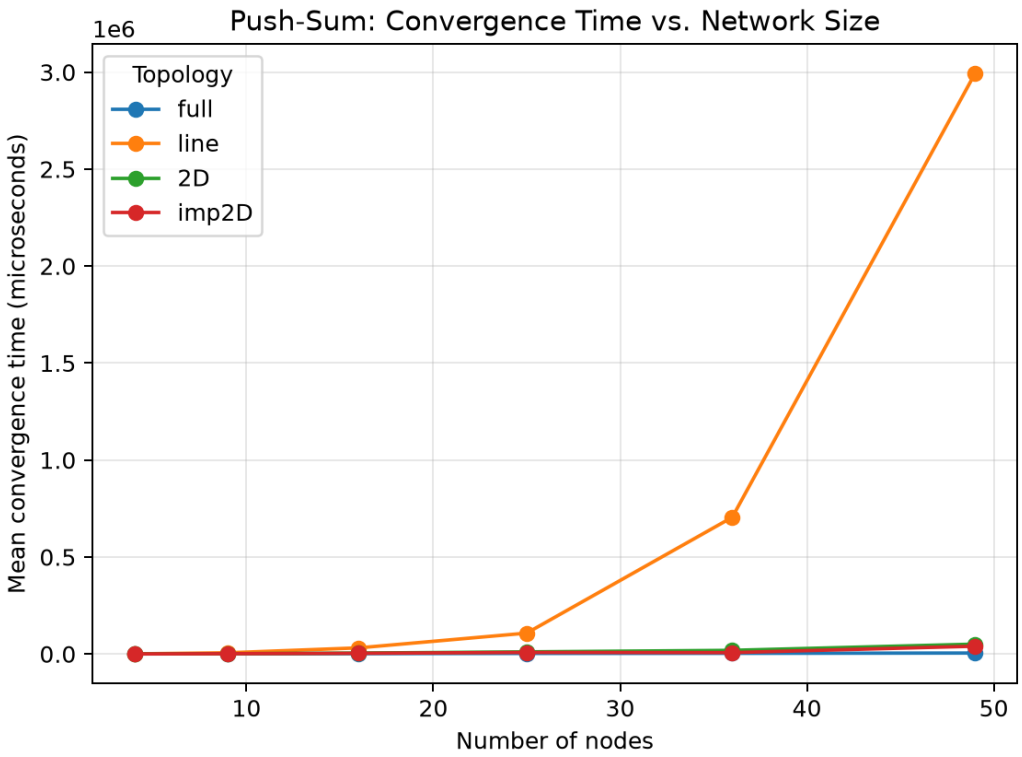


Figure 2: Mean Push-Sum convergence time for successful baseline trials.

Push-Sum's measured convergence time increased substantially with network size in the line topology. At 49 nodes, the mean successful line-topology time was approximately 2.99 seconds (2,994,108 microseconds). For comparison, the corresponding full-topology mean was approximately 4.90 milliseconds (4,901 microseconds). This illustrates the impact that network structure can have on propagation and convergence time in these runs. For Gossip, the baseline threshold of 10 receptions did not produce successful convergence in every configuration. In particular, none of the three baseline trials succeeded for line networks of 25, 36, or 49 nodes. Mean times for configurations with no successful trials are omitted rather than interpreted as zero.

4.3 Largest network with an observed successful trial

The baseline experiments observed at least one successful Gossip trial at the following largest tested network sizes:

Topology	Largest observed successful size
Full	49 nodes
Line	16 nodes
2D	25 nodes
Imperfect 2D	25 nodes

These sizes describe the baseline threshold of 10 and the tested network sizes only. They are not maximum capacities of the implementation. The threshold experiment subsequently observed successful 49-node Gossip runs on every topology at higher thresholds.

5 Gossip Threshold Experiment

5.1 Overall success rate

The additional experiment changes the local reception threshold while keeping the topology, network sizes, and number of trials within the same experimental design. Table 2 shows the aggregate outcomes across all topologies and network sizes.

Table 2: Gossip outcomes by local reception threshold.

Threshold	Successful	Failed	Success rate
10	34	38	47.2
20	56	16	77.8
50	64	8	88.9
100	69	3	95.8
500	71	1	98.6

The observed aggregate success rate increased as the threshold increased. At threshold 10, fewer than half of the trials succeeded. At threshold 500, 71 of 72 trials succeeded. This pattern is consistent with the mechanism of the implementation: actors that reach their local threshold stop forwarding, and a low threshold can allow forwarding to stop before all actors have received the rumor.

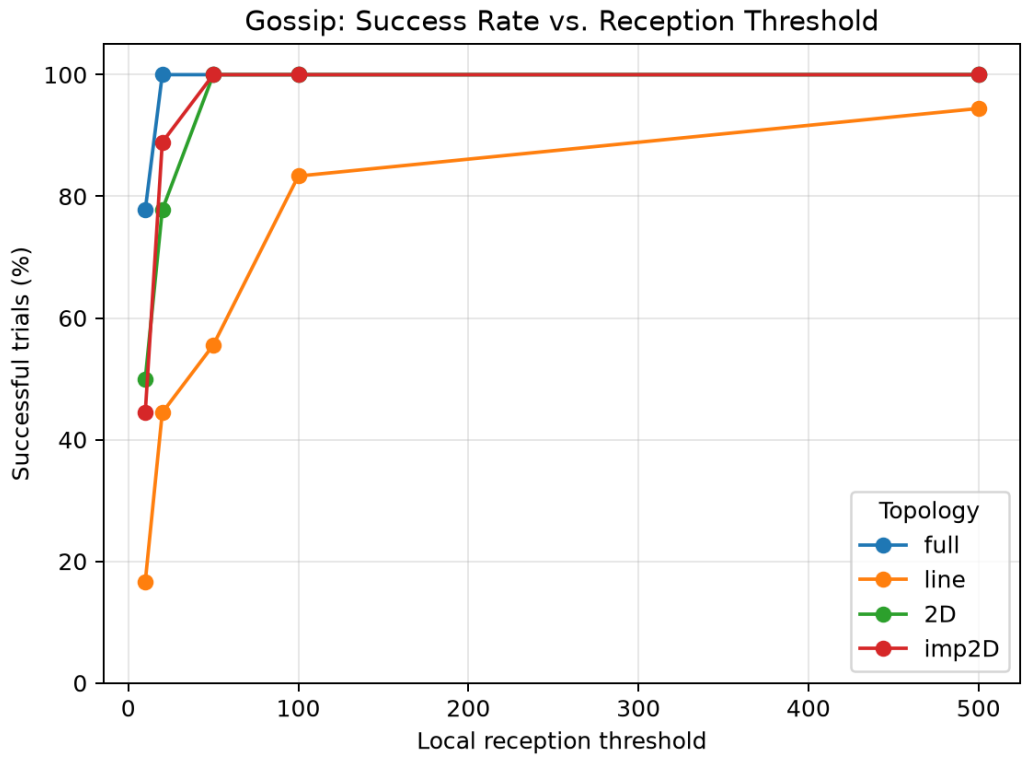


Figure 3: Gossip success rate versus local reception threshold.

5.2 Topology effects

Across all five thresholds and six network sizes, each topology contributes 90 trials. Table 3 summarizes their outcomes.

Table 3: Gossip threshold experiment outcomes by topology.

Topology	Successful	Failed	Success rate
Full	86	4	95.6
Line	53	37	58.9
2D	77	13	85.6
Imperfect 2D	78	12	86.7

The line topology remained the most difficult overall. Its limited neighbor connectivity means that information must travel along the chain, and randomized forwarding can delay propagation or leave parts of the network unreached before actors stop forwarding. The full topology provides many more possible communication paths, which is consistent with its higher observed success rate. The imperfect 2D topology performed slightly better than the regular 2D topology in aggregate in this dataset. The additional random connection may provide shortcuts, although the small number of trials does not establish that this difference will always occur.

5.3 Large line network: follow-up observations

Two additional runs were recorded separately from the 360-trial dataset. For a 49-node line at threshold 100, the run timed out; partial statistics showed rumor-reception reports from only 15 of the 49 actors. At threshold 500, a subsequent run succeeded, with all 49 actors represented in the statistics report and a measured convergence time of 3,686 microseconds. These two observations are illustrative, not a controlled comparison. Gossip is randomized, so one successful run and one failed run cannot by themselves establish a reliable success probability or typical convergence time. They are consistent with the larger threshold experiment, in which the 49-node line had no successes at thresholds 10, 20, or 50, one success at 100, and three successes at 500.

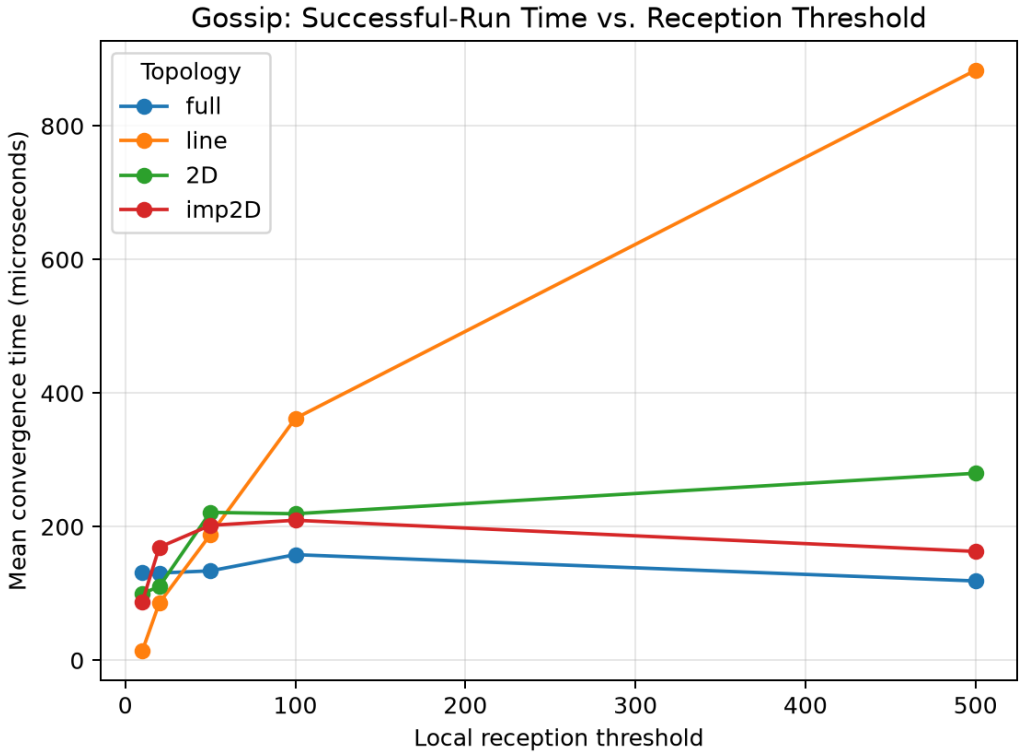


Figure 4: Mean convergence time among successful Gossip trials at each threshold.

The timing graph must be interpreted alongside the success-rate graph. A threshold can have a low mean time among successful runs while still failing frequently, because unsuccessful runs are not included in the mean. Success rate and convergence time therefore provide complementary measures of the experiment.

6 Discussion and Limitations

The experiments support several conclusions about the implemented simulator:

1. **Network topology affects Gossip reliability.** Gossip achieved a higher success rate on the full topology than on the line topology in both the baseline and threshold experiments. In a full network, actors can communicate with many other nodes, whereas a line restricts communication to adjacent nodes.

2. **The reception threshold influences convergence.** Increasing the local Gossip reception threshold was associated with higher aggregate success rates in the tested experiments. A higher threshold allows actors to continue forwarding the rumor for longer before they stop.

3. **Push-Sum succeeded in all baseline trials.** All 72 tested Push-Sum trials converged successfully. This result applies to the tested network sizes, topologies, and execution conditions; it does not guarantee success for larger networks or different configurations.

4. **Convergence time must be interpreted alongside success rate.** The reported mean convergence times include only successful runs. A topology may therefore show a low mean time while also having failed trials. Runtime alone is not sufficient to compare the overall reliability of the algorithms.

The experiments have several limitations. Only three trials were run for each configuration, so randomized execution may affect the observed results. The tested network sizes ranged from 4 to 49 nodes, and the conclusions should not automatically be extended to larger networks. In addition, trials that did not converge within the coordinator's timeout were recorded as failures. A failed trial therefore indicates that convergence was not detected within the configured execution period, rather than proving that convergence is impossible.

Future work could increase the number of trials, test larger networks, record explicit failure reasons, and compare additional stopping criteria. Repeating experiments with fixed random seeds would also improve reproducibility.

7 Conclusion

The Erlang simulator implements asynchronous Gossip and Push-Sum over four network topologies and records convergence statistics. In the baseline experiments, Push-Sum succeeded in all 72 trials, while Gossip succeeded in 39 of 72. Gossip's success was sensitive to topology, with the line network presenting the greatest difficulty. The 360-trial threshold experiment found that aggregate Gossip success increased from 47.2% at threshold 10 to 98.6% at threshold 500. The results demonstrate why the local stopping threshold must be considered when interpreting Gossip convergence. These findings are limited to the tested sizes, three repetitions per configuration, and the current implementation and timeout settings.